

Detailed Marking Report — Mathematics

1st ATTEMPT

exam-12-ch1-2-4-5-7 (Version 1) | Units 1, 2, 4, 5, 7 | Federal Board Annual Examination 2026
Candidate: Seemab | Attempt date: 9 July 2026 | Marked: 9 July 2026 | Fresh paper, first sitting

62/65	95.4%	A+	12/12	31.5/33	18.5/20
TOTAL SCORE	PERCENTAGE	GRADE	SECTION A (MCQ)	SECTION B	SECTION C

All three sections present and clearly readable across 10 scanned pages (CamScanner). This is Seemab's **first attempt** at exam-12-ch1-2-4-5-7 (Units 1, 2, 4, 5, 7), a fresh paper. Every question was attempted — Sections B and C first, then all 12 MCQ answers listed on the final page (allowed). Every deduction below was independently re-verified against the exam's own worked key by a second pass, and the deduction log (3.0 marks lost) reconciles exactly with the headline 62/65.

Section A — Multiple Choice Questions (12 × 1 = 12 Marks)

MCQ answers listed on the last page. Every letter matches the Version 1 key exactly — a clean 12/12.

#	Topic	Key	Seemab	Result	Mark
1	Distributive property: $a(b+c-d)$	B	B	✓	1
2	Number that is its own multiplicative inverse	C	C	✓	1
3	Solve $\log(x+3) = \log(15x-4)$	A	A	✓	1
4	Characteristic of $\log 0.00327$	B	B	✓	1
5	Common factor: $x(2a+3b+7)+x(2a+3b+5)$	A	A	✓	1
6	Add $2a/(a^2-1)$ and $-a/(a^2-1)$	B	B	✓	1
7	Square root of $36x^6y^{16}$	C	C	✓	1
8	Subtract to make 4 a solution of $7x=30$	B	B	✓	1
9	Solution set of $ x +7 = 3$	C	C	✓	1
10	Distance $A(c,10)$ to $B(0,-1) = 20 \rightarrow c$	B	B	✓	1
11	Midpoint of hypotenuse of a right triangle	C	C	✓	1
12	Centre of circle from diameter endpoints	C	C	✓	1

Section A Total	12 / 12
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Perfect 12/12 — 2nd perfect MCQ in a row since the streak reset (Chemistry exam-14). Math MCQ all-time now 95/96 (99.0%).

Section B — Short Questions (11 × 3 = 33 Marks)

Seemab answered the OR variant for 2(i), (iv), (v), (vi), (viii), (ix), (x), (xi); the main part for 2(ii), (iii), (vii). Nine of eleven scored full marks.

Q2(i) OR — Equivalent radicals: (a) $(-27)^{1/3}$ (b) $(\sqrt{16})^4$ (c) $(\sqrt[3]{-8})^9$

3/3

(a) $(-27)^{1/3} = \sqrt[3]{-27} = -3$. **Correct (1/1)** — **Watch:** a stray “or 1/3” was written beside the correct -3 ; commit to a single answer.

(b) $(\sqrt{16})^4 = 4^4 = 256$. **Correct (1/1)**

(c) $(\sqrt[3]{-8})^9 = (-2)^9 = -512$. **Correct (1/1)**

Q2(ii) main — Identify the property

2.5/3

(a) $4 \times 1/4 = 1 \rightarrow$ Multiplicative Inverse property. **Correct (1/1)**

(b) $6 + 0 = 6 \rightarrow$ written “additive property”. **-0.5:** the precise name is the Additive **Identity** property; the word “identity” is what distinguishes it. (0.5/1)

(c) $2 \times (3 \times 5) = (2 \times 3) \times 5 \rightarrow$ Associative property of multiplication. **Correct (1/1)**

Q2(iii) main — Laws of logarithms

3/3

(a) $\log(5pq^2/xy^3) = \log 5 + \log p + 2\log q - \log x - 3\log y$. **Correct (1.5/1.5)**

(b) $3\log x - 5\log y = \log(x^3/y^5)$. **Correct (1.5/1.5)**

Q2(iv) OR — Evaluate $\log^b(0.024)$

3/3

$0.024 = 24/1000 = (2^3 \times 3)/(2^3 \times 5^3) = 3/5^3$. $\log^b(0.024) = \log^b 3 - 3\log^b 5 = 0.4771 - 2.0970 = -1.6199$. *Correct (3/3)*

Q2(v) OR — Factorize by substitution

3/3

Let $u = y^2 + 2y$. $(u-3)(u+11)+48 = u^2+8u+15 = (u+3)(u+5)$. Back-substitute $\rightarrow (y^2+2y+3)(y^2+2y+5)$. *Correct (3/3)*

Q2(vi) OR — Express $8+12t+6t^2+t^3$ as a product of three factors

2/3

Recognised $t^3+6t^2+12t+8 = (t+2)^3 = (t+2)(t+2)(t+2)$. *Correct product (2/2)*

-1: the question also asks to state whether each factor is a binomial or a trinomial — this was left out. Each factor $(t+2)$ is a **binomial**; writing that earns the last mark. (0/1)

Q2(vii) main — Factorize by substitution

3/3

Let $u = 2x^2 - 3x$. $(u+6)u - 55 = u^2+6u-55 = (u+11)(u-5)$. Back-substitute, then factor the second bracket: $(2x^2-3x+11)(2x^2-3x-5) = (2x^2-3x+11)(2x-5)(x+1)$. *Correct (3/3)* — *Note:* a stray “H.C.F” label and a $+3x/-3x$ slip on the very last line only; the established working shows $-3x$ and the answer is right.

Q2(viii) OR — Solve $5 - |5y+1| = -9$

3/3

$|5y+1| = 14 \rightarrow 5y+1 = 14$ or $5y+1 = -14 \rightarrow y = 13/5$ or $y = -3$. Solution set $\{13/5, -3\}$. *Correct (3/3)*

Q2(ix) OR — Solve $4(2y+3) - (6y-1) > 10$

3/3

$8y+12 - 6y+1 > 10 \rightarrow 2y+13 > 10 \rightarrow 2y > -3 \rightarrow y > -3/2$. *Correct (3/3)*

Q2(x) OR — B is the midpoint of AC

3/3

$((3+x)/2, (4+y)/2) = (-1, 7) \rightarrow 3+x = -2 \rightarrow x = -5$; $4+y = 14 \rightarrow y = 10$. *Correct (3/3)*

Q2(xi) OR — P(2,3) midpoint of A(x,4), B(-3,y)

3/3

$(x-3)/2 = 2 \rightarrow x = 7$; $(4+y)/2 = 3 \rightarrow y = 2$. *Correct (3/3)*

Section B Total

31.5 / 33

Section C — Long Questions ($4 \times 5 = 20$ Marks)

Seemab answered the main part for 3(i), (ii); the OR variant for 3(iii), (iv). Three of four fully correct.

Q3(i) main — Evaluate $\log^b(0.0036)$

5/5

$0.0036 = 36/10000 = (2^2 \times 3^2)/(2^4 \times 5^4) = 3^2/(2^2 \times 5^4)$. $= 2\log 3 - 2\log 2 - 4\log 5 = 0.9542 - 0.6020 - 2.7960 = -2.4438$. Positive-mantissa adjustment: $-2.4438 + 3 - 3 = 0.5562 - 3 = \mathbf{3.5562}$ (characteristic -3 , mantissa 0.5562). *Correct throughout (5/5)* — the bar-notation step is exactly right.

Q3(ii) main — Factorize $x^3p^2-8y^3p^2-4x^3q^2+32y^3q^2$

5/5

Group: $p^2(x^3-8y^3) - 4q^2(x^3-8y^3) = (x^3-8y^3)(p^2-4q^2)$. Then $x^3-8y^3 = (x-2y)(x^2+2xy+4y^2)$ and $p^2-4q^2 = (p-2q)(p+2q)$. Result $(x-2y)(x^2+2xy+4y^2)(p-2q)(p+2q)$. *Correct (5/5)* — *Note:* a momentary $+8y^3$ appears in one intermediate line but the correct common factor is extracted on the next line (self-corrected, answer fully right).

Q3(iii) OR — (a) Solve $[\sqrt{(5x-4)}-4]/10 = -1$ (b) Solve $1.3x-0.2 = 0.3x-1.5$

5/5

(a) $[\sqrt{(5x-4)} - 4]/10 = -1 \rightarrow \sqrt{(5x-4)} = -6$. A principal square root can never be negative $\rightarrow$ no real solution, $\emptyset$ (extraneous). *Correct (3/3)*

(b) $1.3x - 0.3x = -1.5 + 0.2 \rightarrow 1.0x = -1.3 \rightarrow x = -1.3$. *Correct (2/2)*

Q3(iv) OR — $\triangle A(3,7)$, $B(-4,0)$, $C(4,0)$; D, E midpoints of AB, AC

3.5/5

(a) D = midpoint AB = $(-0.5, 3.5)$; E = midpoint AC = $(3.5, 3.5)$. *Correct (1.5/1.5)*

(b) $d(D,E) = \sqrt{[(4)^2+0^2]} = 4$; $d(B,C) = \sqrt{[(8)^2+0^2]} = 8$. *Correct (2/2)*

(c) *-1.5:* the comparison and geometric conclusion were not written. Since $d(B,C) = 2 \cdot d(D,E)$, DE is parallel to BC and equal to half its length (Midsegment / Midpoint Theorem). Both distances AND this relationship are needed for the last 1.5 marks. (0/1.5)

Section C Total

18.5 / 20

Final Score Breakdown

Section	Available	Earned	%	Notes
Section A — MCQ	12	12	100%	All 12 correct; Math MCQ all-time 95/96 (99.0%)
Section B — Short Questions	33	31.5	95.5%	Lost 1.5: Q2(ii)(b) property name (0.5), Q2(vi) binomial statement (1.0)
Section C — Long Questions	20	18.5	92.5%	Lost 1.5: Q3(iv)(c) midsegment conclusion not stated
TOTAL	65	62	95.4%	Grade A+

Deduction Log (all 3.0 marks lost)

Question	Sub-part	Issue	Lost
Q2(ii)	(b) $6+0=6$	Named “additive property”; the precise name is the Additive Identity property (word “identity” missing).	−0.5
Q2(vi) OR	classify factors	Three-factor product $(t+2)(t+2)(t+2)$ correct, but did not state each factor is a binomial.	−1.0
Q3(iv) OR	(c) conclusion	$d(D,E)=4$ and $d(B,C)=8$ both correct, but the comparison $d(B,C)=2 \cdot d(D,E) / DE \parallel BC$ (midsegment) was never stated.	−1.5

Strengths in this attempt

- **Perfect MCQ (12/12)** across all four units — 2nd perfect in a row since the streak reset; Math MCQ record now 95/96 (99.0%).
- **Logarithms fully mastered:** log expansion, $\log^b(0.024) = -1.6199$, and the hard bar-notation evaluation $\log^b(0.0036) \rightarrow 3.5562$ all correct, including the positive-mantissa adjustment.
- **Factorization strong:** both substitution factorizations and the four-factor grouping $x^3p^2-8y^3p^2-4x^3q^2+32y^3q^2 = (x-2y)(x^2+2xy+4y^2)(p-2q)(p+2q)$ all correct.
- **Radical equation understood, not guessed:** recognised $\sqrt{5x-4} = -6$ has no real solution ($\emptyset$) rather than pushing through mechanically.
- Absolute-value equation, compound inequality, and every coordinate-geometry midpoint problem (2(x), 2(xi), 3(iv) computations) clean and fully correct.

Areas to revise

- **Write the final naming / conclusion statement — this is the whole story of the 3 lost marks.** Every deduction is the same kind: the arithmetic is right, but the required closing line is missing.
 - Q2(ii)(b): write “Additive **Identity**”, not just “additive property”.
 - Q2(vi): after $(t+2)(t+2)(t+2)$, add “each factor is a **binomial**”.
 - Q3(iv): after finding 4 and 8, add “ $d(B,C) = 2 \cdot d(D,E)$, so $DE \parallel BC$ and is half its length (midsegment theorem)”.
- **This exact habit cost marks on the exam-11 papers too** (dropped rectangle conclusion, missing counterexample). Making “always state the concluding sentence / name the property / classify the answer” automatic would put her at essentially full marks — the computation is not the problem.
- **Minor:** avoid two competing answers on one line (the stray “or 1/3” in 2(i)(a), the “H.C.F” mislabel in 2(vii)). Commit to one boxed final answer.